

Mortal Computation

Mortality, the Genome, and Life-History in Cost-Accounted GSLTs

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Abstract

We read the cost-accounting (metering) construction on graph-structured lambda theories (GSLTs) as a biology rather than as an economics. The token-gated rewrite that meters computation is, on the nose, a mechanism for making computation *mortal*: death is deadlock-by-starvation, and metabolism is the splitter/joiner token economy that postpones it. For Turing-complete continued-interactive GSLTs the metered calculus translates faithfully back into the original model, and the image of the round trip is a sublanguage in which mortality and metabolism are implemented from native parts. Turing completeness further yields, by composing the two universality maps, a meta-circular interpreter; when the model hosts reflection natively this interpreter is internalised, giving the quote/dequote pair the dual-use character von Neumann required of a self-reproducing description. Quoted description behaves as a potentially immortal germ line; the dequoted running process is the mortal soma. With reproduction thereby isolated as a metabolic cost separable from a computation's other functions, we recover life-history theory: the logistic law $\dot{N} = rN(1 - N/K)$ emerges as the mean-field of a population of cost-accounted replicators on a bounded-inflow substrate, with r and K as closed forms in the construction's primitives, and r - versus K -selection appears as the two extreme vertices of a continuous allocation simplex.

1 Setup: the cost endofunctor and its round trip

Write an interactive GSLT as $\mathcal{G} = (\Sigma, E, R)$ with a distinguished interacting sort carrying a labelled transition system, structural equations E , and rewrite rules R . The cost construction is an endofunctor $\mathbf{C}$ on the subcategory of interactive GSLTs and bisimulation-preserving morphisms. On objects $\mathbf{C}$ adjoins a signature sort $\mathbf{Sig}$ as a commutative monoid, a token-stack sort $S ::= () \mid s : S$ as the free $\mathbf{Sig}$ -graded structure, a signed-term wrapper $\{\cdot\}_s$, and a constructor $\mathbf{sig-of}$ that mints a fresh signature from runtime data. It replaces each rule $\ell \rightarrow r$ of R by its *gated family*: the redex fires only when matching tokens are present and consuming them. The five rules that govern signed communication are not five facts; they are the image of the single communication rule under $\mathbf{C}$.

We assume throughout that $\mathcal{G}$ is Turing-complete and *continued-interactive*: it can express persistent processes that continue to interact after producing output. Under this assumption the metered calculus $\mathbf{C}\mathcal{G}$ admits a compositional translation

$$\mathbf{T} : \mathbf{C}\mathcal{G} \longrightarrow \mathcal{G},$$

which is a faithful simulation up to bisimulation: signatures map to channel names (via reflection where available, or to fresh names otherwise), tokens become messages on those channels carrying continuation payloads, and signed processes become *fuel-gates*—guard patterns that must

consume a token message before the underlying redex proceeds. Splitters and joiners reconcile combined and decomposed token granularities. The object of study here is the *round trip*

$$\mathsf{T} \circ \mathsf{C} : \mathcal{G} \longrightarrow \mathcal{G},$$

an endomorphism of the original model. Its image $\text{im}(\mathsf{T} \circ \mathsf{C}) \subseteq \mathcal{G}$ is a distinguished sublanguage built entirely from native constructs. The claim of this note is that the biology—mortality, the genome, and life-history—lives in that image, and that all of it is therefore native to any Turing-complete continued-interactive $\mathcal{G}$, the cost construction merely locating where.

2 Mortality and metabolism

Death. In CG a redex fires only by consuming a matching token; a process whose stack is $()$ has no enabled rewrite and blocks permanently. Pushing this through T , the token message on the relevant signature-channel simply stops arriving and the fuel-gate never completes. In $\mathcal{G}$ this is an ordinary configuration: a guarded process whose guard is never discharged.

Definition 1 (Starvation death). A process P in $\text{im}(\mathsf{T} \circ \mathsf{C})$ is *dead* at a configuration if every redex of P is fuel-gated and no token message is reachable on any of its gating channels. Death is permanent deadlock by fuel exhaustion.

Mortality is thus not a new primitive but a fragment of $\mathcal{G}$ that $\mathcal{G}$ admitted all along; C exhibits which fragment.

Metabolism. The token economy that postpones death is metabolic in a precise sense. A joiner assembles a high-denomination token from components (anabolism); a splitter cleaves one into spendable units (catabolism); and the gated rewrite couples each unit of computational work to a unit of fuel consumption—no step without an accompanying expenditure. This is the same coupling that defines a metabolism thermodynamically. Read against a Ruelle–Bowen potential [9, 1] on the transition graph, the splitter/joiner dynamics are the flow and the zero-fuel configurations are the absorbing set: metabolism is motion on the landscape, death is its absorbing boundary.

The round-trip image and the Buss argument. A parallel composition of metered processes is exactly the kind of population that supports selection: copies are autonomous ($P \mid P \not\equiv P$ once embedded in distinct local contexts), and selection exploits inexactness rather than collapsing it. In the un-metered $\mathcal{G}$ an immortal replicator $!P$ with a faster-cycling P always invades a parallel composition without bound; there is no configuration in which a higher-level whole is stable against its own fastest subterm. This is Buss’s cell-lineage problem [2] in its rawest form. In $\text{im}(\mathsf{T} \circ \mathsf{C})$ the defector is fuel-bounded: it out-cycles its neighbours only until its stack is spent, after which it deadlocks.

Proposition 1 (Mortality caps invasion). *Let D be a subterm-variant with per-cycle fuel draw strictly exceeding its per-cycle fuel income on a shared bounded substrate. In CG , D executes for a number of cycles bounded by its initial endowment and then dies, so the cumulative interactions it contributes are finite. In the un-metered $\mathcal{G}$ the same D contributes unboundedly. Hence configurations in $\text{im}(\mathsf{T} \circ \mathsf{C})$ that sequester the minting constructor sig-of admit no heritable takeover by a high-draw somatic defector, whereas no configuration of un-metered $\mathcal{G}$ resists one.*

The biological reading is Buss’s: somatic mortality together with germ sequestration aligns the levels of selection by denying a within-individual defector any future. Mortality is selected not in spite of bounding individual runs but because a bounded run is the precondition for a stable higher-level individual that can itself be a durable unit of selection.

3 Reflection and the genome

A self-interpreter from universality. Turing completeness furnishes two universality maps. The first, $e : \text{TM} \rightarrow \mathcal{G}$, is completeness proper: $\mathcal{G}$ simulates any machine. The second, $i : \mathcal{G} \rightarrow \text{TM}$, is the Church–Turing direction [11]: any effectively presented reduction relation can be run by a machine. Their composite

$$i \circ e : \mathcal{G} \longrightarrow \mathcal{G}$$

is a program *in* $\mathcal{G}$ that takes a $\mathcal{G}$ -program as data and runs it, correct up to bisimulation: a meta-circular interpreter. To feed a program to this interpreter one must first represent it as a value, and the TM detour manufactures such a code-as-data representation with no reflective primitive in sight. *Reflection appears as a consequence of universality*—a theorem, not a term. This is exogenous reflection.

What native reflection changes. When $\mathcal{G}$ hosts reflection natively—as the rho calculus does [6], a reflective higher-order extension of Milner’s π -calculus [7], where $@P$ is already a value in the same algebra as the running process—the meta-circular interpreter is internalised, and the gain is representational, not merely a saved detour. A TM-tape encoding has a brutal mutation neighbourhood: almost any edit to the tape is catastrophic because the tape does not respect the term structure it encodes. A quoted term $@P$ has a benign one: mutation is a structural edit of a subterm and recombination is the splicing of quoted subprocesses, both staying inside the model’s own grammar. Native reflection is therefore the difference between a genome that is evolvable and one that is not.

Proposition 2 (Two readings agree on behaviour, differ on the description). *The exogenous interpreter $i \circ e$ and the internalised interpreter agree up to bisimulation on the behaviour of every interpreted program, but differ on the algebra of the code-as-data representation: the former carries a foreign encoding closed under tape edits, the latter the model’s own term algebra closed under structural edits and splices. Evolvability is a property of the latter algebra, not of the interpreted behaviour, and so distinguishes the two readings even though they are behaviourally equivalent.*

Von Neumann’s dual use, recovered. A self-reproducing description φ [12] must support two distinct operations: it is *copied uninterpreted* (replication) and *interpreted to build the machinery* (expression). Native reflection supplies both on a single object. A name $@P$ can be transmitted on a channel and received *without* dequoting—replication, the description copied as inert data—or dequoted by $*$ and run—expression, the description read as instructions. Crick’s transcription/translation split [3] and Weismann’s two modes of the germ-line description [14] are these same two operations on $@P$. The meta-circular structure is what closes the loop: the reader is encoded in what it reads.

Germ and soma from quote and dequote. Refining §2: the germ line is the sublanguage that retains **sig-of** *and* carries the quoted description $@P$ —together exactly what is needed to fuel and to instantiate an offspring. The soma is the dequoted running process $*@P$. Weismann’s mortality asymmetry then falls out of the quote/dequote distinction rather than being postulated:

$$\underbrace{*@P}_{\text{soma}} \text{ metabolises and starves; } \quad \underbrace{@P}_{\text{germ}} \text{ is inert, cannot starve, is re-instantiable.}$$

Mortal soma = the dequoted process. Immortal germ = the quoted description.

Finally, applying a fuel-gate not to a bare communication but to the *dequoting* operation is regulated expression: the gene $@P$ is read and run only when the requisite tokens are present. This is operon logic—expression coupled to metabolic state. The genome’s full operation set is thus native: replication is copying the name, expression is token-gated dequoting, mutation and recombination are edits and splices of the quoted term, and regulation is the fuel-gate deciding *when* the description is read.

4 Life-history theory

Metering makes “the cost of reproduction” a measurable quantity rather than a category error. In un-metered $\mathcal{G}$, $!P$ replicates for free and nothing is separable from anything else. Once every operation is token-gated, the reproduction operations—minting via **sig-of**, copying $@P$, endowing the offspring’s initial stack, and the gated dequoting that instantiates it—each carry an identifiable charge in the same currency as every other charge the process incurs.

4.1 The metabolic ledger

Per metabolic cycle a replicating computation has realised income B and partitions it into maintenance and reproduction,

$$B = c_{\text{maint}} + c_{\text{rep}}, \quad c_{\text{rep}} \approx n(c_{\text{mint}} + c_{\text{copy}} + e),$$

where n is the number of offspring per cycle and e is the *endowment*—the initial token stack that travels with each copied description $@P$. The endowment is literally metabolic: tokens removed from the parent’s budget and placed in the offspring’s. The partition (maintenance versus reproduction, and within reproduction n versus e) is the life-history allocation problem stated in the construction’s own primitives.

4.2 Carrying capacity as a fuel-balance fixed point and the emergence of the logistic law

Let a bounded token inflow Φ (tokens per unit time) feed the substrate through the splitter/joiner network, shared among N identical replicators. Realised per-process income is density-dependent through contention on the bounded inflow. To leading order we take the contention response to be linear,

$$B(N) = B_0 \left(1 - \frac{N}{N_{\text{sat}}} \right), \quad N_{\text{sat}} = \frac{\Phi}{B_0}, \quad (1)$$

the first-order term of any saturating response that vanishes when aggregate demand NB_0 exhausts the inflow. Per-capita birth rate is the reproductive surplus divided by the unit cost of reproduction, and death has a baseline density-independent rate d_0 (accident and exogenous shock) on top of starvation:

$$b(N) = \frac{\max(0, B(N) - c_{\text{maint}})}{c_{\text{rep}}}, \quad d(N) = d_0. \quad (2)$$

Proposition 3 (Emergence of the logistic law). *Under (1)–(2), the mean-field population dynamics of a set of cost-accounted replicating computations on a bounded-inflow substrate, in the regime $B(N) \geq c_{\text{maint}}$, obey the logistic law [13]*

$$\frac{dN}{dt} = (b(N) - d_0) N = rN \left(1 - \frac{N}{K}\right),$$

with intrinsic rate and carrying capacity given in closed form by the ledger primitives,

$$r = \frac{B_0 - c_{\text{maint}}}{c_{\text{rep}}} - d_0, \quad K = N_{\text{sat}} \left(1 - \frac{c_{\text{maint}} + d_0 c_{\text{rep}}}{B_0}\right) = \frac{\Phi}{B_0} \left(1 - \frac{c_{\text{maint}} + d_0 c_{\text{rep}}}{B_0}\right).$$

Derivation. Substituting (1) into (2) and dropping the clamp in the stated regime,

$$b(N) - d_0 = \frac{B_0(1 - N/N_{\text{sat}}) - c_{\text{maint}}}{c_{\text{rep}}} - d_0 = \underbrace{\left(\frac{B_0 - c_{\text{maint}}}{c_{\text{rep}}} - d_0\right)}_r - \frac{B_0}{c_{\text{rep}} N_{\text{sat}}} N.$$

Multiplying by N gives $\dot{N} = rN - (B_0/(c_{\text{rep}} N_{\text{sat}}))N^2$, which is $rN(1 - N/K)$ with $K = r c_{\text{rep}} N_{\text{sat}}/B_0$. Expanding r and using $N_{\text{sat}} = \Phi/B_0$ yields the stated closed form. $\square$

Remark. The quadratic term—hence the entire logistic shape—comes from the density-dependence of realised income, i.e. from contention on the bounded inflow. The law is therefore *derived*, not assumed. The qualitative content—carrying capacity set by inflow over per-capita draw, intrinsic rate set by surplus over reproductive unit cost—is robust to the precise contention ansatz; the exact coefficient in K depends on it. In the maintenance-dominated small- d_0 limit the carrying capacity reduces to the bare fuel balance $N^* c_{\text{maint}} = \Phi$, i.e. $K \rightarrow \Phi/c_{\text{maint}}$: as many processes as the inflow can keep from starving. Density-dependent mortality is not imposed; it emerges as the rising starvation rate when $N c_{\text{maint}}$ approaches Φ and the splitters can no longer top every stack.

4.3 r - and K -selection as corners of an allocation simplex

Selection acts on the allocation, and which regime is favoured is set by which term in the ledger binds; the r/K dichotomy [5, 8] is the coarsest reading of it. The criterion is the invasion analysis of §2; only the binding constraint differs.

- **r -corner.** Uncrowded or volatile substrate ($N \ll K$, death dominated by exogenous supply shocks, density-independent) rewards maximising $r = (B_0 - c_{\text{maint}})/c_{\text{rep}} - d_0$: push n up, push e and c_{maint} down. Many minimal-endowment offspring, fire-and-forget, short-lived soma. This is the immortal-replicator defector of §2, now adaptive rather than pathological, because in an empty substrate raw instantiation rate is what invasion rewards.
- **K -corner.** Saturated, stable substrate ($N \approx K$, death dominated by competition, density-dependent) rewards competitive fuel capture: push e and c_{maint} up, push n down. Richly-endowed offspring, long-lived competitive soma; per-offspring capture beats raw count.

Corollaries in the primitives. Parental care is a persistent process that keeps feeding tokens to the offspring’s signature-channel after instantiation—a continued-interactive contract, which is what “continued” is for; r -strategists cut the channel at birth, K -strategists hold it open. Semelparity is the parent dumping its residual budget into one reproductive burst and then deadlocking by design ($c_{\text{maint}} \rightarrow 0$ post-burst); iteroparity retains maintenance to survive further cycles. Body size maps to the soma’s standing token stack—the capital a process keeps on hand—and the familiar r/K body-size correlation appears because large standing capital repays its maintenance draw only when competition, not throughput, binds.

Remark (The construction gives more than the dichotomy). r and K are the two extreme vertices of a continuous allocation simplex over $(c_{\text{maint}}, n, e, \text{timing})$. That continuum is exactly modern age-structured / disposable-soma life-history theory [10, 4], which r/K coarsens into a binary. The construction natively produces the continuum, so the right computational taxonomy of reproduction schemes is a classification of points in the simplex with r and K as named corners—truer to the biology and more natural in the construction than a dichotomy.

5 What to make rigorous next

Three load-bearing claims are stated above at the level a working note allows and deserve full treatment.

1. **Idempotence of the round trip up to bisimulation.** If $(T \circ C)^2 \approx T \circ C$, then $\text{im}(T \circ C)$ is exactly the set of bisimulation fixed points of the metabolism comonad—the processes already mortal and metabolising—and “the organisms inside $\mathcal{G}$ ” acquires a crisp definition. The entire biological reading rests on this image being a well-defined sublanguage rather than a construction artifact.
2. **The fidelity comparison (Prop. 2).** Make precise the sense in which the exogenous and internalised self-interpreters agree on behaviour up to bisimulation while differing on the description algebra, and prove that evolvability—benign mutation neighbourhood under structural edits—is a property of the description algebra alone.
3. **The logistic-emergence proposition (Prop. 3).** Promote the mean-field derivation to a theorem: specify the stochastic token-flow process whose hydrodynamic limit is the stated ODE, identify the conditions on the splitter/joiner network under which the contention response (1) is the correct leading order, and characterise the corrections that distinguish the exact K from the bare fuel balance Φ/c_{maint} . This is the technical spine of the programme.

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